

Independent evolution of a living bridge in the Old World army ant lineage

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Abstract

Living bridges formed by army ants are among the most striking examples of collective behavior in social insects, previously known only from New World *Eciton* species. Here, we report the observation of living bridge formation in an Old World army ant, *Aenictus*. During field observations, workers formed bridges across small gaps on foraging trails by linking their bodies, allowing traffic to resume. Bridges were maintained while traffic was high but spontaneously disassembled as flow declined, suggesting a similar behavioral algorithm to that of *Eciton* self-assembly behavior. This finding indicates that bridge-formation has evolved convergently in army ants and provides new insight into the diversity of self-assembly in social insects.

Keywords:

Collective motion, Natural history, self-organization, traffic regulation, tropical biology

Introduction

Collective behavior enables a group of animals to perform tasks that single individuals cannot accomplish (Camazine et al., 2001). One striking example is self-assembly, in which individuals physically link their bodies to create dynamic and flexible three-dimensional structures on the spot (Anderson et al., 2002; Carlesso and Reid, 2023). Such structures provide functional advantages, otherwise impossible for single individuals, such as forming temporary shelters, bridging gaps to facilitate movements, or combining forces to achieve greater strength. Self-assembly is rare across animals and is largely restricted to several derived social Hymenopteran species, e.g., fire ant rafts, honey bee swarms, weaver ant chains, and army ant bivouacs (reviewed in (Carlesso and Reid, 2023)). Although these are system-specific phenomena, similar ecological pressures may promote the convergent evolution of self-assembly behaviors across unrelated taxa.

Army ants represent a particularly charismatic example of collective behavior, exhibiting a suite of traits known as the “army ant syndrome”, including obligate group foraging and nomadic colony cycles (Gotwald, 1995; Kronauer, 2020; Schneirla, 1971). This syndrome has evolved convergently in two different lineages: the New World army ants and the Old World army ants (Borowiec, 2019). Among these, the genera *Eciton* (New World) and *Aenictus* (Old World) are especially surface-adapted, typically foraging and migrating on open ground, rather than within leaf litter or subterranean tunnels (Gotwald, 1995). As terrestrial animals, army ant movements depend on the continuity of the ground surface, potentially disrupted by the small gap (Mizumoto and Reid, 2024). The New World army ant, *Eciton*, is well known for its ability to overcome such spatial discontinuities through collective behavior. These ants form living bridges that span gaps, and the behavioral mechanisms have been well documented (McCreery et al., 2022; Reid et al., 2015). In contrast, in the Old World army ant, *Aenictus*, behavioral and life history information remain scarce. Different from other army ant groups, *Aenictus* does not have polymorphic workers and forms a more simplified collective foraging compared to *Eciton* ants (Gotwald, 1995; Schneirla,

1971), yet their surface-adapted lifestyle suggests that they might also employ similar bridging behaviors.

Here, we report observations of living bridges in *Aenictus* ants. This finding provides rare insight into collective behavior in the Old World army ants and suggests that comparable ecological challenges may have driven the convergent evolution of self-assembly behaviors across independent lineages of army ants.

Methods

All observations were performed at the Lambir Hills National Park, Miri, Sarawak, Malaysia. Between May 31 and June 12, 2023, we conducted daily field surveys of marching termites (*Hospitalitermes* spp. and *Longipeditermes longipes*) along the ~4 km trail (Inoue-Pantu-Main trail), walking the route once or twice per day to search for the termite trail formations. During these surveys, we occasionally encountered foraging or emigration trails of the Old World army ants, *Aenictus* spp., both during the day and at night. On June 6, while returning to the chalet after observations on *Hospitalitermes lividipes*, we encountered the tail of an *Aenictus* ant at 10:20 pm local time (4.2023437° N, 114.0302110° E). The trail crossed a narrow twig of dead wood on the forest floor, providing a natural opportunity to test whether this species could form a living bridge when the substrate was interrupted.

We cut the twig bridge with scissors and recorded the ant responses using the 4K video camera (HC-X1500-K, Panasonic) at 30 fps, illuminated by an LED video light (Pixel G1s). The resulting gap was about 5.3 mm wide, slightly larger than the body length of ants. Preliminary observations indicated that LED lights at night did not disturb ant activity on the trail. Recording lasted for about 400 s, limited by battery capacity and safety considerations for returning from the field site.

To quantify traffic dynamics during bridge formation, we analyzed the video using the open-source event-logging software BORIS (Friard and Gamba, 2016). The observation area was defined as a 50 × 50 mm square centered on the twig gap (Fig. S1). We counted the number of ants entering and leaving the area each second and binned the data accordingly. The traffic flow was defined as the number of individuals entering or leaving the area per second, and the estimated number of ants present was derived from cumulative entry and exit counts over the observation period (410 s total). These data were used to describe the temporal dynamics of collective bridge formation.

The autocorrelation function of the traffic flow data shows that there was no detectable correlation with a lag $\geq$ of 11 seconds. Thus, we summarized the data for every 11 seconds by getting the mean of the traffic flow and the number of ants near the gap. Then, we investigated the effect of the number of ants near the gap, bridge formation, and their interactions on the traffic flow, using a linear model (LM). All analyses were performed using R (R Core Team, 2024).

Results

During the observation period, we observed three bridge formation events and two spontaneous collapses (Fig. 1, Video S1). After the gap was created in the middle of the trail on a twig, traffic became congested, and ants accumulated near the opening. As the accumulation increased, some individuals were pushed toward the gap and began exploring the air space with their legs and antennae (Fig. 1). When ants from opposite sides made contact, they interlinked their bodies to form a living bridge. Once the bridge was established, traffic flow resumed, and the local density of ants around the gap decreased. The bridge persisted as long as ants continued to cross, but collapsed spontaneously when traffic declined (Fig. 1).

During bridge formation, the minimum number of ants present in the gap area was seven, indicating that at least seven individuals were required to span the 5.3 mm gap. The number of ants near the gap was positively correlated with traffic flow (linear model: $F_1 = 45.3$, $P < 0.001$,

Fig. S2), and the slope of this relationship was significantly steeper when the bridge was present ($F_1 = 5.0$, $P = 0.032$, Fig. S2).

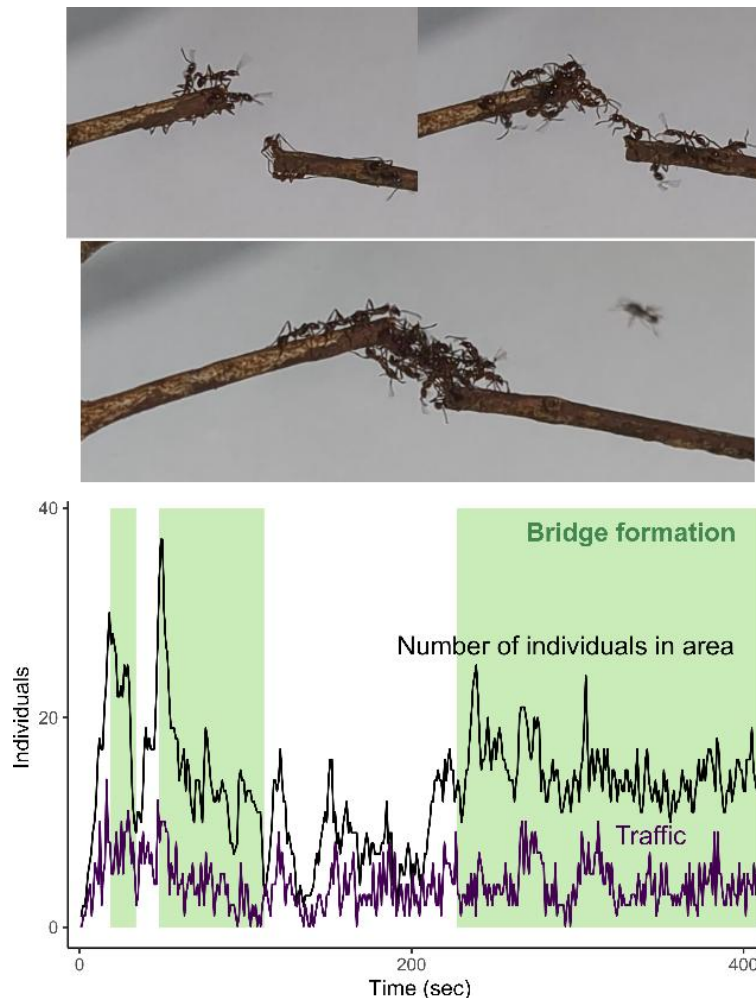


Figure 1. Living bridge formation by *Aenictus* sp. Time development of the number of individuals around the gap (25 mm) was recorded. Also, traffic was measured as the number of individuals who enter or leave this area per second.

Discussion

Old World and New World army ants show many convergent traits, yet also display key differences in their collective behaviors. Both *Aenictus* and *Eciton* are surface-adapted foragers (Gotwald, 1995), but *Eciton* species typically form larger and more complex raid networks, while *Aenictus* raids tend to be simpler (Gotwald, 1995; Schneirla, 1971). Our observations reveal that *Aenictus* ants can also form living bridges as in *Eciton*, indicating that this behavior evolved convergently. However, bridge formation in *Aenictus* appears more limited, typically spanning only small gaps and often breaking down spontaneously when traffic decreases (Fig. 1). This likely reflects the lower overall traffic intensity in *Aenictus* trails (Schneirla and Reyes, 1966), as in our observed trail. Observations on larger-scale trials could further clarify the extent of this behavior.

Living bridge formation in *Eciton* workers can emerge from simple behavioral rules. Individual ants track the rate of traffic passing over them and leave the bridge when tactile stimulation drops below a certain threshold (Garnier et al., 2013; McCreery et al., 2022; Powell and Franks, 2007), which could be used in other contexts as well (Baudier and Pavlic, 2020; Lutz et al., 2021). Our

observations suggest that *Aenictus* employs a similar mechanism. We recorded two spontaneous bridge collapses coinciding with declines in local traffic and ant density (Fig. 1). This could be consistent across other lineages (e.g., (Mlot et al., 2011)). Among self-assembly behaviors in social Hymenoptera, bivouac formation is the most widespread, providing a temporary shelter for colony members (Anderson et al., 2002; Carlesso and Reid, 2023). Both Old World and New World army ants form bivouacs, including several *Aenictus* species (Gotwald, 1995). We also observed clustering behavior at our field site (Fig. S3). The underlying behavioral rules that govern bivouac and bridge formation are likely related, suggesting that *Aenictus* may have co-opted a common self-assembly mechanism for different contexts.

In conclusion, we document the first evidence of living bridge formation in Old World army ants. Although this behavior is well known in *Eciton*, its evolutionary origin remains unclear. Unique foraging behavior in army ants, such as mass raiding, is suggested to have continuously evolved from group foraging (Chandra et al., 2021). Yet, behavioral information within the subfamily Dorylinae remains scarce. Our findings underscore the continuing importance of natural history observations in revealing behavioral diversity and convergence among social insects.

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Author contributions

N.M.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, validation, visualization, writing-original draft

C.H.K.: resources, supervision

T.K.: investigation, writing-edit

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